

PART I: CONCEPT-WISE QUESTION BANK

Jadavpur University Previous Year Questions (2022 - 2025)

1. Operating Principles & Kinematics

- [2022, Q1(a)] Discuss the similarities between a transformer and an induction machine. Hence, explain why an induction machine is called a generalized transformer. (2022)[cite: 1]
- [2024, Q1(a)] Why an induction machine is called a generalized transformer? (2024)[cite: 3]
- [2022, Q1(b)] What will be the direction of rotation of the rotor, if the primary winding is placed on the rotor of a poly-phase induction motor? (2022)[cite: 1]
- [2024, Q1(b)] What will be the direction of rotation of the rotor, if the primary winding is placed on the rotor of a poly-phase induction motor? (2024)[cite: 3]
- [2025, Q1(b)] What will be the direction of rotation of the rotor, if the primary winding is placed on the rotor of a poly-phase induction motor? Give reasons for your answer. (2025)[cite: 4]
- [2022, Q1(c)] Explain why the rotor of an induction motor can never attain synchronous speed while the rotor field with respect to the stator rotates with the synchronous speed. (2022)[cite: 1]
- [2023, Q1(a)] What is the speed of rotor mmf of a 3-phase induction motor with respect to its stator mmf? - Justify your answer. (2023)[cite: 2]
- [2024, Q1(c)] What is the relative speed between the magnetic field produced by stator winding and the magnetic field produced by the rotor winding? Give reasons for your answer. (2024)[cite: 3]
- [2025, Q1(c)] What is the relative speed between the magnetic fields produced by the stator winding and rotor winding of a 3-phase induction motor? (2025)[cite: 4]
- [2025, Q1(a)] "It is not absolutely necessary to laminate the rotor of an induction motor." True or False? Give reasons. (2025)[cite: 4]
- [2025, Q1(e)] Describe the phenomenon of cogging and crawling. (2025)[cite: 4]

2. Stator Windings, RMF & Space Harmonics

- [2022, Q2(a)] Prove that, a three phase balanced supply on a balanced three phase winding can produce a rotating magnetic field with constant magnitude and with synchronous speed. (2022)[cite: 1]
- [2023, Q2(a)] "A rotating magnetic field of constant amplitude can be produced by supplying a balanced three phase voltage source to a balanced three phase winding" Explain. Find out the speed of the rotating magnetic field produced. How can you change the direction of the rotating magnetic field? (2023)[cite: 2]
- [2024, Q2(a)] Explain how a rotating magnetic field with constant amplitude is generated in an three phase induction machine. (2024)[cite: 3]
- [2025, Q2(a)] Explain how a poly-phase winding fed from a balanced poly-phase supply produces a rotating magnetic field with constant amplitude. (2025)[cite: 4]
- [2023, Q1(b)] Why stators in a 3-phase induction motor (except large rating ones) are usually provided with semi-closed type slots? Explain. (2023)[cite: 2]
- [2022, Q1(d)] What do you mean by distributed winding? Find an expression for distribution factor. (2022)[cite: 1]
- [2024, Q1(d)] Find an expression for pitch factor for a short pitched winding. (2024)[cite: 3]
- [2025, Q1(d)] Find an expression for distribution factor of a distributed winding. (2025)[cite: 4]
- [2023, Q1(e)] Show that a 3-phase winding carrying sinusoidal currents produces space harmonics of the order of $h = 6k \pm 1$, where k is a positive integer. (2023)[cite: 2]
- [2023, Q1(f)] Show that, in an three phase induction motor, the 5^{th} harmonic flux produces backward rotating torque at a speed of $1/5^{th}$ synchronous speed whereas 7^{th} harmonic produces forward rotating torque at a speed of $1/7^{th}$ synchronous speed. (2023)[cite: 2]

3. Equivalent Circuit & Parameter Testing

- [2022, Q3(a)] Draw and explain the phasor diagram of a three phase induction motor and develop the equivalent circuit. (2022)[cite: 1]
- [2023, Q3(a)] Develop the equivalent circuit of a three phase induction motor and explain how the mechanical power developed is taken care in the equivalent circuit. (2023)[cite: 2]
- [2024, Q3(a)] From the basic principle develop the equivalent circuit of a poly phase induction motor. (2024)[cite: 3]
- [2025, Q3(a)] Develop the equivalent circuit of an induction motor from the basic principle. (2025)[cite: 4]
- [2022, Q3(b)] Describe different tests to determine the equivalent circuit parameters of a three phase induction motor. How can you calculate the equivalent circuit parameters from these test results? (2022)[cite: 1]
- [2023, Q3(b)] Describe different tests to determine the equivalent circuit parameters of a three phase induction motor. How can you calculate the equivalent circuit parameters from these test results? (2023)[cite: 2]
- [2024, Q3(b)] How can you determine the equivalent circuit parameters in the laboratory? (2024)[cite: 3]
- [2025, Q3(b)] How the equivalent circuit parameters of an induction motor are determined in the laboratory? (2025)[cite: 4]

4. Torque, Power & Performance Relationships

- [2022, Q2(b)] Derive an expression for torque developed in an induction motor. Draw the torque-speed characteristics. (2022)[cite: 1]
- [2023, Q2(b)] Derive an expression for torque developed in an induction motor and draw the torque-speed characteristics. At what slip the torque developed in an induction motor will be maximum? (2023)[cite: 2]
- [2024, Q2(a)] Derive an expression for developed torque of an induction motor. Also derive the value of slip for which torque is maximum and draw the torque-speed characteristics. (2024)[cite: 3]
- [2025, Q2(b)] Derive an expression for power developed in an induction motor. Also derive the condition for maximum power and draw the power-speed characteristics. (2025)[cite: 4]
- [2022, Q1(e)] The maximum torque of a three phase induction motor does not depend on the rotor resistance - Explain. (2022)[cite: 1]
- [2023, Q1(d)] What is the relationship of developed torque of a 3-phase induction motor with the supply voltage? Justify your answer. (2023)[cite: 2]
- [2022, Q4(a)] i) If the stator impedance of an induction motor is neglected show that: $T_e/T_{em} = 2/(s_{mT}/s + s/s_{mT})$ ii) From the equivalent circuit of a poly phase induction motor, obtain the following relations: 1) $I_{2st}/I_2 = \sqrt{[(s^2 + s_{mT}^2)/(s^2(1 + s_{mT}^2))]}$ 2) $I_{2mT}/I_2 = \sqrt{[0.5(1 + (s_{mT}/s)^2)]}$. (2022)[cite: 1]
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5. Starting Methods, Deep Bar & Double Cage Rotors

- [2022, Q1(f)] What is deep bar rotor induction motor? How it produces high starting torque? (2022)[cite: 1]
- [2023, Q1(g)] What is deep bar rotor induction motor? How it produces high starting torque? (2023)[cite: 2]
- [2024, Q1(e)] What is deep bar rotor induction motor? How it produces high starting torque? (2024)[cite: 3]
- [2025, Q1(f)] What is deep bar rotor induction motor? How it produces high starting torque? (2025)[cite: 4]
- [2022, Q4(b)] Describe the construction of a double cage induction motor and explain how high starting torque is developed in double cage induction motor. Draw the equivalent circuit of double cage rotor induction motor. (2022)[cite: 1]
- [2024, Q4(b)] Describe a double cage induction motor. Draw and explain the equivalent circuit of double cage rotor induction motor. (2024)[cite: 3]
- [2023, Q4(b)] Why starters are necessary for starting an induction motor? What are the various types of starters used for starting of squirrel cage induction motor? Describe with circuit diagram the working of any one type of starter for starting squirrel cage induction motor and hence derive an expression for starting torque in terms of full-load torque. (2023)[cite: 2]
- [2025, Q4(a)] Describe different starting methods of a 3-phase induction motor. (2025)[cite: 4]
- [2025, Q4(b)] What are the advantages and disadvantages of a SRIM over a SQIM? (2025)[cite: 4]

6. Numerical Problems

- **[2022, Q5(a)]** A 440 V, 3-phase, 4-pole, 50 Hz, 37.3 kW, Y-connected induction motor has the following parameters: $R_1 = 0.1 \, \Omega$, $X_1 = 0.4 \, \Omega$, $R'_2 = 0.15 \, \Omega$, $X'_2 = 0.44 \, \Omega$. Motor has stator core loss of 1250 W and rotational loss of 1000 W. It draws a no-load line current of 20 A at a power factor of 0.09 (lag). When motor operates at a slip of 3%, calculate (i) input line current and power factor, (ii) electromagnetic torque developed in N-m, (iii) output and (iv) efficiency of the motor. (2022)[cite: 1]
- **[2022, Q5(c)]** A 25 kW, 6-pole, 50 Hz, 400 V, 960 rpm, efficiency 90%, power factor 0.86 SCIM takes 200 amperes on direct on-line starting. Determine the ratio of starting to full-load torque and starting to full-load current in case of (a) DOL starting, (b) star-delta starting, (c) auto-transformer starting with 80% tapplings and (d) series resistance starting limiting the starting current to 120 amperes. (2022)[cite: 1]
- **[2023, Q1(c)]** An 8-pole, 3-phase, 60° spread, double layer winding has 72 coils in 72 slots. The coils are short-pitched by two slots. Calculate the winding factor for the fundamentals and third harmonics. (2023)[cite: 2]
- **[2023, Q5(a)]** The power input to a 6-pole, 50 Hz, 3-phase induction motor is 700 W at no-load and 10 kW at full-load. The no-load copper losses may be assumed negligible while the full-load stator and rotor copper losses are 295 W and 310 W respectively. Find the full-load speed, shaft torque and efficiency of the motor assuming the rotational and core losses to be equal. (2023)[cite: 2]
- **[2023, Q5(b)]** The resistance and reactance (equivalent) values of a double-cage induction motor for stator, outer and inner cage are 0.25, 1.0 and 0.15 ohm resistance and 3.5, zero and 3.0 ohm reactance respectively. Find the starting torque if the phase voltage is 250 V and the synchronous speed is 1000 rpm. (2023)[cite: 2]
- **[2024, Q5(a)]** The rotor resistance and standstill reactance of a 3-phase induction motor are 0.012 ohm and 0.08 ohm per phase respectively. The full load slip is 4% at normal voltage. Calculate the percentage reduction in stator voltage to develop the full-load torque at 75% of full-load speed. (2024)[cite: 3]
- **[2024, Q5(b)]** The standstill impedances of the two cages of a double-cage induction motor are $(3.2 + j1.2) \, \Omega$ and $(0.5 + j6.5) \, \Omega$ respectively. If the full-load slip is 5%, then find the ratio of starting torque to full-load torque. Neglect magnetizing current and stator impedance. (2024)[cite: 3]
- **[2025, Q5(a)]** A 50 kW, 400 volt, 3-phase, 4-pole, 50 Hz wound rotor induction motor develops a maximum internal torque of 240% at a slip of 20% when working under rated voltage and frequency conditions and with slip ring short-circuited. Determine (a) full load slip, (b) rotor copper loss at full load and (c) starting torque in N-m. Neglect stator resistance and rotational losses. (2025)[cite: 4]
- **[2025, Q5(b)]** A 15 kW, 3-phase, 4 pole, 50 Hz, induction motor has a full load slip of 0.03. Mechanical and stray load losses at full load are 3.5% of output power. Compute i) Power delivered by stator to rotor ii) Electromagnetic torque at full load iii) Rotor ohmic loss at full load. (2025)[cite: 4]